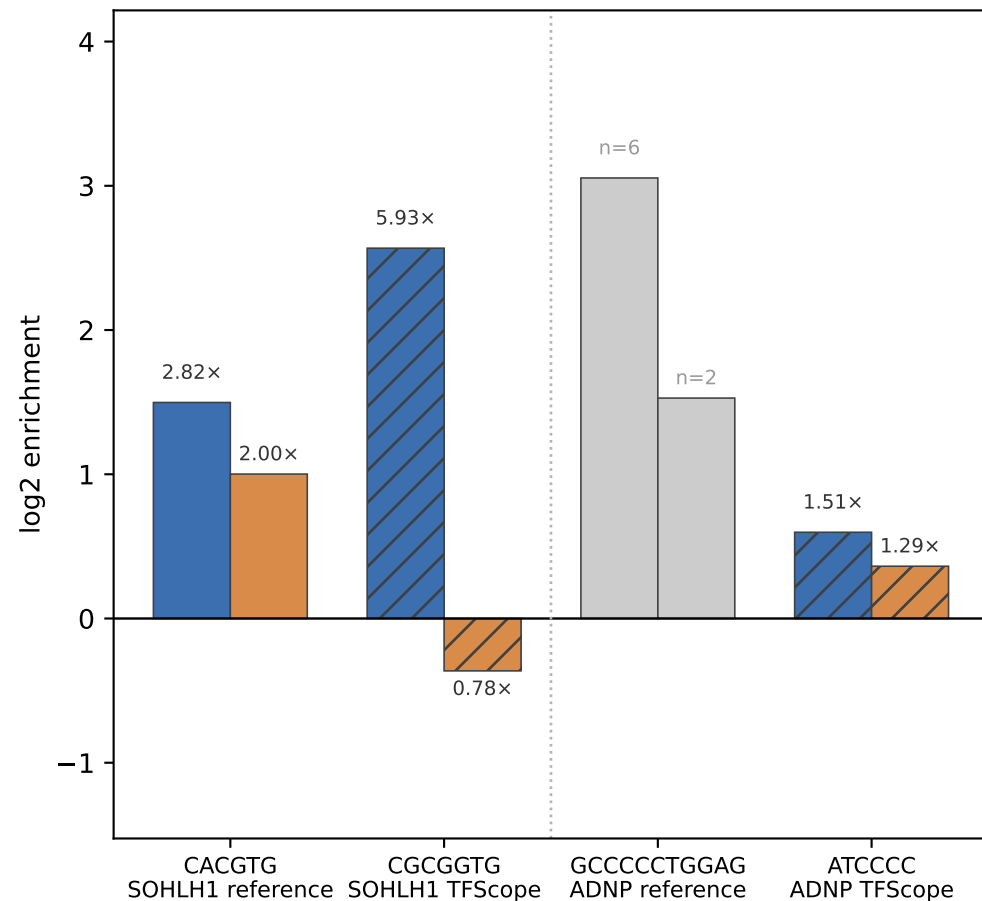


Exact-consensus cCRE enrichment, with the experimental motif as a positive control

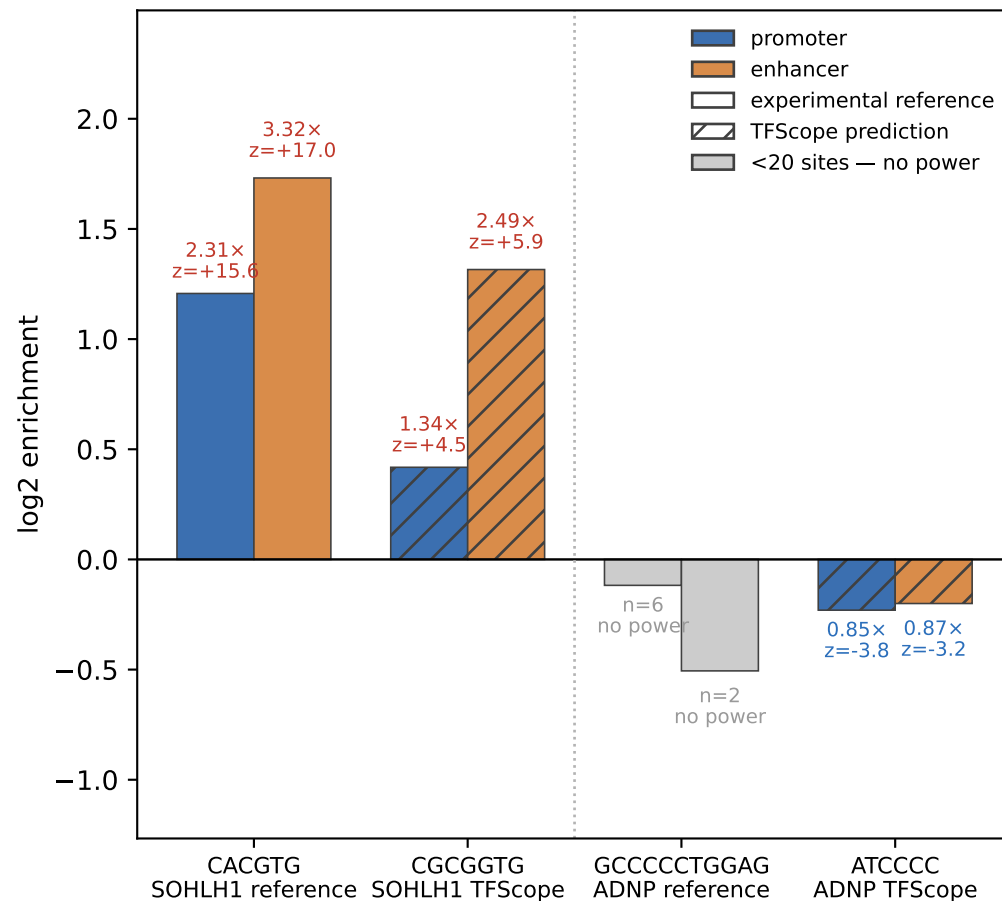
b Naive: cCREs vs whole genome

hit density in cCREs / hit density across hg38 — confounded by GC content



c Composition-controlled: cCREs vs dinucleotide shuffle

observed / mean of 100 exact Altschul-Erikson dinucleotide shuffles of the same cCREs



The validated E-box CACGTG is strongly enriched (2.31x promoter, 3.32x enhancer) — the assay detects a real motif.
TFscope's CGCGGTG is enriched in the same direction but at lower amplitude (1.34x / 2.49x): a blurred CACGTG, not a random GC word.
ADNP's reference GCCCCTGGAG occurs only 6x (promoter) and 2x (enhancer), so the positive control has no power and ADNP cannot be tested by this assay; its predicted ATCCCC is a generic 6-mer with 1,024,921 genomic sites. Empirical p floors at 1/101 → z is the discriminator.